

Mario Kart Score Distributions

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1 Introduction

In a 12-player Mario Kart race, each player is awarded a certain number of points based on their placement in that race. Currently, Mario Kart games use the following point distribution (in placement-descending order):

$$\begin{bmatrix} 15 \\ 12 \\ 10 \\ 8 \\ 7 \\ 6 \\ 5 \\ 4 \\ 3 \\ 2 \\ 1 \\ 0 \end{bmatrix}$$

But is this the best point distribution? It depends. It is impossible to universally define what “best” means here. One could define the best point distribution system to be the one that, when used identically for each race, best predicts the outcomes of the entire n -race competition. One could also define “best” to be the one that most accurately ranks the skill of the racers. Even then, skill is not a quantifiable measure. Similarly, for the formerly-proposed definition, we create a point distribution to determine the outcomes of an n -race competition, so asking which one best “predicts” the outcome brings us straight back to our first question: what is the best point distribution system?

The essential problem rooted within this question is that an identical placement-based point distribution must be used for every race of every n -race competition, meaning that the point system may not use information relating to the players’ finishing times, the players’ positions at times in the race other than at the finish line, etc.

In this article I formulate some alternatives to the games’ point distribution system based on techniques involving Markov chains.

2 Using Markov Chains

If the reader is not familiar with the concept of Markov chains, they will be briefly explained here as we will be using them, though this explanation is far from a comprehensive explanation of Markov chains.

Let P_k denote the player that came in k th place.

We will treat the players as nodes. Each player gets 1 “vote” to distribute between the other players based on the place that they came in. This is represented by a matrix, which we will denote as M , for which the entry in the i th row and j th column (i.e. M_{ij}) is the portion of P_j ’s vote that goes toward P_i . After the first round of “voting,” all of the players get to vote with however many votes they received from the last round. We can raise M to a very high power to determine how many votes each player has after many, many rounds of voting.

Here, we want players to cast more votes to the players that finish ahead of them than to those who finish behind them. There are many ways to accomplish this. Our first Markov chain approach will use the ratios of the placements of each pair of players.

To make it simpler to compare the point distributions, we will formulate each one so that 12th place gets 0 points and so that the sum of all the points distributed is equal to that of the Mario Kart games:

$$15 + 12 + 10 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 + 0 = 73.$$

Additionally, after initially calculating the votes that each player gives every other player, we need to be careful to scale each column so that each player gets exactly 1 vote.

2.1 Ratios of Placements

Here,

$$M_{ij} = \frac{j}{i},$$

e.g. P_9 gives P_4 $\frac{9}{4}$ votes and each player gives themselves 1 vote, which gives us the matrix:

$$\begin{bmatrix} \frac{1}{1} & \frac{2}{1} & \frac{3}{1} & \cdots & \frac{12}{1} \\ \frac{1}{2} & \frac{2}{2} & \frac{3}{2} & \cdots & \frac{12}{2} \\ \frac{1}{3} & \frac{2}{3} & \frac{3}{3} & \cdots & \frac{12}{3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{12} & \frac{2}{12} & \frac{3}{12} & \cdots & \frac{12}{12} \end{bmatrix}.$$

Now, we must scale each column by 1 over its sum to ensure that each player gets exactly 1 “vote,” which produces the following matrix (each entry is rounded to the nearest thousandth):

[	0.322	0.322	0.322	0.322	0.322	0.322	0.322	0.322	0.322	0.322	0.322	0.322	]
[	0.161	0.161	0.161	0.161	0.161	0.161	0.161	0.161	0.161	0.161	0.161	0.161	]
[	0.107	0.107	0.107	0.107	0.107	0.107	0.107	0.107	0.107	0.107	0.107	0.107	]
[	0.081	0.081	0.081	0.081	0.081	0.081	0.081	0.081	0.081	0.081	0.081	0.081	]
[	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064	]
[	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	0.054	]
[	0.046	0.046	0.046	0.046	0.046	0.046	0.046	0.046	0.046	0.046	0.046	0.046	]
[	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	]
[	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	0.036	]
[	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	0.032	]
[	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	0.029	]
[	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	0.027	]

Notice that each column is identical (which makes sense: the value of $\frac{j}{\sum_{i=1}^{12} \binom{i}{i}}$ does not depend on j), which means that raising M to a any power would yield itself since M is stochastic. This gives us the point distribution:

$$\begin{bmatrix} 0.322 \\ 0.161 \\ 0.107 \\ 0.081 \\ 0.064 \\ 0.054 \\ 0.046 \\ 0.04 \\ 0.036 \\ 0.032 \\ 0.029 \\ 0.027 \end{bmatrix}.$$

Now, we subtract 0.027 from each entry in this vector and scale it so that the sum of all points distributed is 73. This yields the final distribution of points, with each score rounded to the nearest tenth for ease of readability:

$$\begin{bmatrix} 31.8 \\ 14.5 \\ 8.7 \\ 5.8 \\ 4.0 \\ 2.9 \\ 2.1 \\ 1.4 \\ 1.0 \\ 0.6 \\ 0.3 \\ 0 \end{bmatrix}.$$

Notice that this point distribution is much more heavily skewed towards the higher placements than the one used in-game.

2.2 Ratios of Placement to Sum of Placements

Suppose we want to distribute the initial votes so that each “pair” of votes - that is P_a ’s quantity of votes towards P_b and vice versa - contributes “equally” to the final point distribution. For example, in the last section, the votes between P_1 and P_{12} summed to $12 + \frac{1}{12}$ - much more than the sum of the votes between P_1 and P_2 .

Let us entertain this idea of “equal contribution” by making each pair of votes sum to 1. We can accomplish this by having

$$M_{ij} = \frac{j}{i+j}$$

since it follows from this that $M_{ij} + M_{ji} = 1$ for all i, j where $i \neq j$. For entries on the diagonal (when $i = j$), $M_{ij} = \frac{1}{2}$, i.e. each player gives half a vote to themselves. After scaling each column of M to make M column stochastic, we get:

[	0.229	0.19	0.168	0.154	0.144	0.137	0.131	0.126	0.122	0.119	0.117	0.114	]
[	0.153	0.143	0.135	0.128	0.124	0.12	0.116	0.114	0.111	0.109	0.108	0.106	]
[	0.115	0.114	0.112	0.11	0.108	0.106	0.105	0.103	0.102	0.101	0.1	0.099	]
[	0.092	0.095	0.096	0.096	0.096	0.096	0.095	0.095	0.094	0.094	0.093	0.093	]
[	0.076	0.082	0.084	0.086	0.086	0.087	0.087	0.087	0.087	0.088	0.087	0.087	]
[	0.066	0.071	0.075	0.077	0.079	0.08	0.081	0.081	0.082	0.082	0.082	0.083	]
[	0.057	0.063	0.067	0.07	0.072	0.074	0.075	0.076	0.077	0.077	0.078	0.078	]
[	0.051	0.057	0.061	0.064	0.067	0.068	0.07	0.071	0.072	0.073	0.074	0.074	]
[	0.046	0.052	0.056	0.059	0.062	0.064	0.065	0.067	0.068	0.069	0.07	0.071	]
[	0.042	0.048	0.052	0.055	0.058	0.06	0.062	0.063	0.064	0.066	0.067	0.068	]
[	0.038	0.044	0.048	0.051	0.054	0.056	0.058	0.06	0.061	0.063	0.064	0.065	]
[	0.035	0.041	0.045	0.048	0.051	0.053	0.055	0.057	0.058	0.06	0.061	0.062	]

Once we raise M to a high power and fit the steady state vector so that its 12th entry is 0 and so that the sum of its entries is 73, we get:

$$\begin{bmatrix} 19.5 \\ 14.0 \\ 10.5 \\ 8.1 \\ 6.3 \\ 4.8 \\ 3.7 \\ 2.7 \\ 1.9 \\ 1.2 \\ 0.5 \\ 0 \end{bmatrix}.$$

2.3 Ratios of Placement to Geometric Mean of Placements

The last point distribution could be formulated using the ratios of a racer’s placement to the arithmetic mean of each pair’s placements, since each entry in M would have simply doubled, which would have no bearing on the final result (since we scale each column anyway).

So, now let us try using the geometric mean instead of the arithmetic mean of each pair of racer's placements to quantify how much each "pair" of votes contributes to the final results. This is analogous to using the magnitude of a vector instead of the sum of its components to quantify its size.

Here,

$$M_{ij} = \frac{j}{\sqrt{i^2+j^2}}.$$

This ensures that $(M_{ij})^2 + (M_{ji})^2 = 1$ for all i, j where $i \neq j$. Here $M_{ij} = \frac{1}{\sqrt{2}}$ when $i = j$.

This gives us $M =$

[	0.26	0.196	0.161	0.14	0.127	0.118	0.111	0.106	0.103	0.1	0.097	0.096	]
[	0.164	0.155	0.141	0.129	0.12	0.113	0.108	0.104	0.101	0.098	0.096	0.095	]
[	0.116	0.122	0.12	0.115	0.111	0.107	0.103	0.1	0.098	0.096	0.094	0.093	]
[	0.089	0.098	0.102	0.102	0.101	0.099	0.097	0.096	0.094	0.093	0.092	0.091	]
[	0.072	0.081	0.087	0.09	0.091	0.092	0.091	0.091	0.09	0.09	0.089	0.089	]
[	0.06	0.069	0.076	0.08	0.083	0.084	0.085	0.086	0.086	0.086	0.086	0.086	]
[	0.052	0.06	0.067	0.072	0.075	0.078	0.079	0.081	0.082	0.082	0.083	0.083	]
[	0.046	0.053	0.059	0.065	0.068	0.072	0.074	0.076	0.077	0.078	0.079	0.08	]
[	0.041	0.048	0.054	0.059	0.063	0.066	0.069	0.071	0.073	0.074	0.076	0.077	]
[	0.037	0.043	0.049	0.054	0.058	0.061	0.064	0.067	0.069	0.071	0.072	0.074	]
[	0.033	0.039	0.045	0.049	0.053	0.057	0.06	0.063	0.065	0.067	0.069	0.071	]
[	0.031	0.036	0.041	0.046	0.05	0.053	0.057	0.059	0.062	0.064	0.066	0.068	]

Once we raise M to a high power and fit the steady state vector so that its 12th entry is 0 and so that the sum of its entries is 73, we get:

$$\begin{bmatrix} 18.0 \\ 13.7 \\ 10.6 \\ 8.4 \\ 6.6 \\ 5.1 \\ 3.9 \\ 2.9 \\ 2.0 \\ 1.2 \\ 0.6 \\ 0 \end{bmatrix}.$$

Notice that this distribution is very similar to the last, the difference being that the points here are distributed slightly more evenly.

3 Conclusions and Further Lines of Questioning

Similar lines of reasoning and mathematical techniques could be used more generally for analyzing point distributions in systems of ranking that involve Elo or a similar measure of score that is meant to represent a player's skill, or more generally an item's importance.

This article uses relatively arbitrary methods of formulating a point distribution. Different Markov-chain-related approaches could be applied.

Interestingly the second and third point distributions derived in this article (the ones that use ratio-to-sum and ratio-to-geometric-mean) fit excellently with curves derived through cubic regression, with $R^2 = 0.9976$ and $R^2 = 0.9990$ respectively using the rounded data.

Perhaps a better way to approach this problem would be to use probability distributions of players' times as a function of perceived skill over the course of n races to identify what point distribution most often accurately ranks the players in terms of their skill from the placement data of only one or two races.